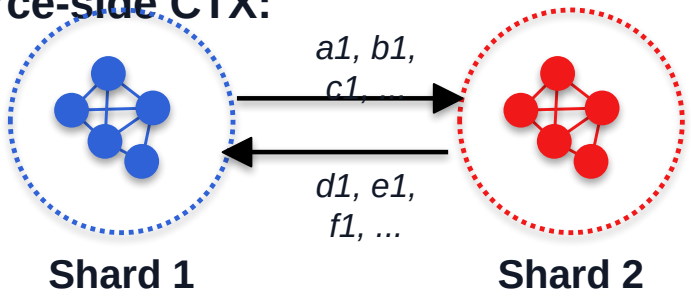
Figure 1. Bilateral Netting Mechanism

Shard 1 nodes are blue; Shard 2 nodes are red. Reciprocal source-side CTXs are offset before destination-side execution.

Shard 1 node

Shard 2 node

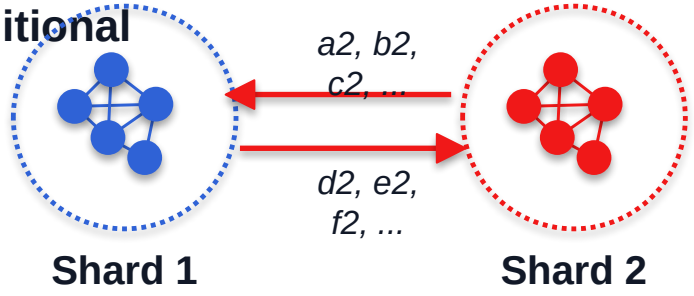
Source-side CTX:



Observation

k reciprocal source-side CTXs are generated between Shard 1 and Shard 2.

Destination-side CTX: Traditional



Traditional

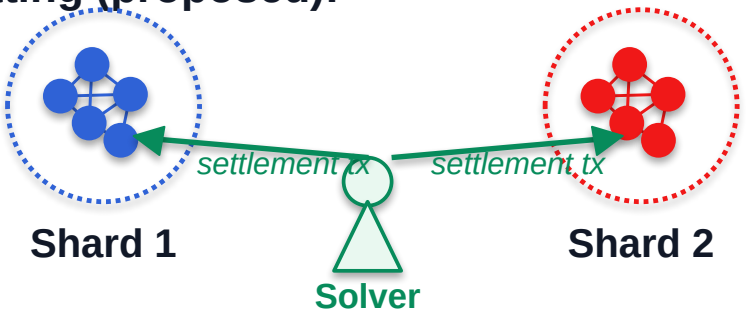
Still processes k destination-side CTXs through PBFT and block packing.

Mechanism insight

Netting offsets reciprocal source-side CTXs before destination-side execution.

Only a compact settlement set (a << k) reaches consensus.

Netting (proposed):



Proposed Netting

Offsets reciprocal CTXs first; only compact settlement CTXs remain (a << k).